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Is $f(x) = (2x)/(x-6)$ continuous at 4?



To determine if the function $f(x) = (2x)/(x-6)$ is continuous at $x = 4$, we need to check three conditions:

1. The function must be defined at $x = 4$.
2. The limit of the function as x approaches 4 must exist.
3. The limit of the function as x approaches 4 must be equal to the value of the function at $x = 4$.

Let's evaluate each condition:

1. The function $f(x) = (2x)/(x-6)$ is defined for all $x \neq 6$. Therefore, it is defined at $x = 4$.
2. To find the limit as x approaches 4, we can substitute 4 into the function:

$$\begin{aligned}\lim_{x \rightarrow 4} (2x)/(x-6) \\ &= (2 * 4)/(4 - 6) \\ &= 8/(-2) \\ &= -4\end{aligned}$$

3. Now, let's evaluate the value of the function at $x = 4$:

$$\begin{aligned}f(4) &= (2 * 4)/(4 - 6) \\ &= 8/(-2) \\ &= -4\end{aligned}$$

Since the limit of the function as x approaches 4 (-4) is equal to the value of the function at $x = 4$ (-4), we can conclude that the function $f(x) = (2x)/(x-6)$ is continuous at $x = 4$.



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